CPSC 304 Project Cover Page

Milestone #: 2

Date: July 30, 2026

Group Number: 41

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

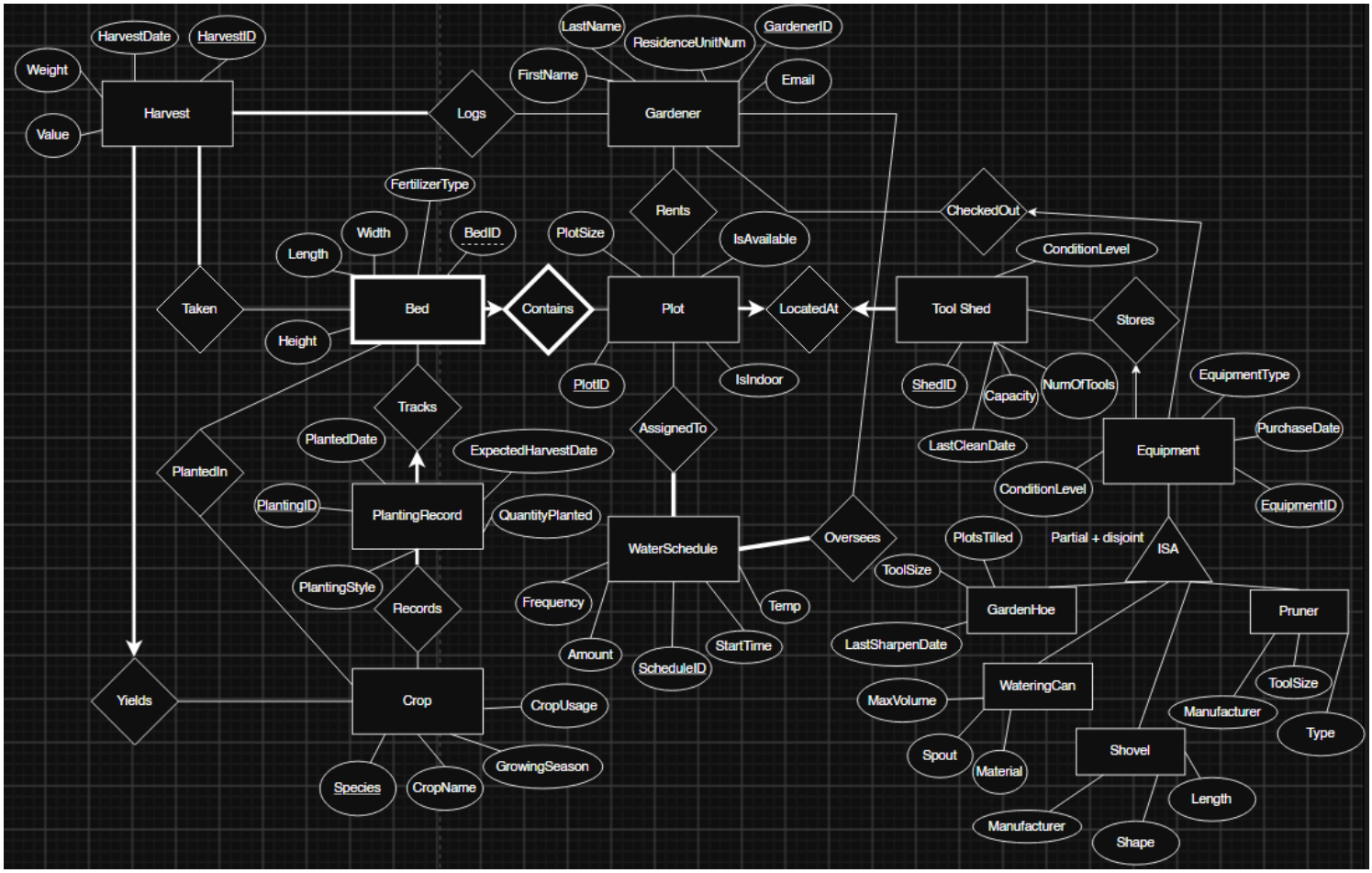
2. A brief (~2-3 sentences) summary of your project. Many of your TAs are managing multiple projects so this will help them remember details about your project.

Our project manages community gardens. It uses a database to allow users to see which gardeners have rented which plot and what equipment, and to assign themselves to plot and equipment. It allows users to create a watering schedule and track the progress of a crop from when it was planted to when it was harvested, including its location, overseer, and the equipment used.

3. State which database and programming language you wish to use for the project (e.g., Javascript/React and the department provided Oracle database). You will not be able to make changes to this decision after this point.

We will be using Javascript, HTML, and CSS for the frontend, Java/JDBC for the backend, and Oracle for DBMS.

4. The ER diagram you are basing your item #3 (below) on.



Changes made:

1) Changed the relationship name between Harvest and Crop from Contains to Yields, reducing duplicate names
2) Changed the relationship name between Tool Shed and Equipment from Contains to Yields, reducing duplicate names
3) Changed the attribute of Equipment from Equipment Brand to ConditionLevel, since the Manufacturer attribute is seen in some subtype entities
4) Changed all names with spaces to UpperCamelCase, to keep consistency
5) Changed Last Inspection attribute in Tool Shed to ConditionLevel, since it was similar to LastCleanDate and to add an FD
6) Changed AvailabilityStatus attribute in Plot to IsAvailable to match boolean naming convention
7) Changed Yields from a many-to-many to a many-to-one relationship between Harvest and Crop, to add an FD and to ensure each harvest is of a specific crop
8) Changed relationship name between Harvest and Bed from From to Taken because of SQL issues
9) Changed relationship name between PlantingRecord and Crop from Of to Records because of SQL issues
10) Changed attribute in Pruner from Size to ToolSize because of SQL issues
11) Changed attribute in Plot from Size to PlotSize because of SQL issues
12) Changed attribute in GardenHoe from Size to ToolSize because of SQL issues

5. The schema derived from your ER diagram (above). For the translation of the ER diagram to the relational model, follow the same instructions as in your lectures. The process should be reasonably straightforward. For each table:

Primary Key: Underline
Foreign Key: Bolded
Candidate Key: Italicized

1. HarvestYields (HarvestID: INT, HarvestDate: DATE, Weight: INT, Value: INT, **Species: VARCHAR**)
    a. HarvestDate, Weight, Value, and Species are NOT NULL, FK (Species) REFERENCES Crop(Species)

2. Gardener (FirstName: VARCHAR, LastName: VARCHAR, ResidenceUnitNum: INT, GardenerID: INT, Email: VARCHAR)
    a. FirstName, LastName, and ResidenceUnitNum are NOT NULL, Email is UNIQUE

3. Plot (PlotSize: INT, IsAvailable: BOOLEAN, PlotID: INT, IsIndoor: BOOLEAN)
    a. IsAvailable is NOT NULL

4. WaterSchedule (Frequency: INT, Amount: INT, ScheduleID: INT, StartTime: VARCHAR, Temp: INT)
    a. Frequency, Amount, and StartTime are NOT NULL

5. PlantingRecordTracks (PlantedDate: DATE, PlantingID: INT, PlantingStyle: VARCHAR, QuantityPlanted: INT, ExpectedHarvestDate: DATE, **BedID: INT**, **PlotID: INT**)
    a. BedID and PlotID are both UNIQUE and NOT NULL, FK(BedID, PlotID) REFERENCES Bed(BedID, PlotID)

6. Crop (Species: VARCHAR, CropName: VARCHAR, GrowingSeason: VARCHAR, CropUsage: VARCHAR)
    a. CropName and GrowingSeason are NOT NULL

7. ToolShedLocatedAt (ShedID: INT, LastCleanDate: DATE, Capacity: INT, NumTools: INT, ConditionLevel: INT, **PlotID: INT**)
    a. Capacity and NumTools are NOT NULL, PlotID is NOT NULL and UNIQUE, FK(PlotID) REFERENCES Plot(PlotID)

8. StoresAndCheckedOutEquipment (EquipmentID: INT, PurchaseDate: DATE, EquipmentType: VARCHAR, ConditionLevel: INT, **ShedID: INT**, **GardenerID: INT**)
    a. PurchaseDate and EquipmentType are NOT NULL, FK (ShedID) REFERENCES ToolShedLocatedAt(ShedID), FK (GardenerID) REFERENCES Gardener(GardenerID)

9. BedContains (Length: INT, Width: INT, Height: INT, FertilizerType: VARCHAR, BedID: INT, **PlotID: INT**)
    a. Length, Width, and Height are NOT NULL, PK (BedID, PlotID), FK (PlotID) REFERENCES Plot(PlotID)

ISA:

10. GardenHoe (**EquipmentID: INT**, PlotsTilled: INT, ToolSize: INT, DateLastSharpened: DATE)
    a. ToolSize is NOT NULL, FK (EquipmentID) REFERENCES StoresAndCheckedOutEquipment (EquipmentID)

11. WateringCan (**EquipmentID: INT**, MaxVolume: INT, Spout: VARCHAR, Material: VARCHAR)
    a. MaxVolume, Spout, and Material are NOT NULL, FK (EquipmentID) REFERENCES StoresAndCheckedOutEquipment (EquipmentID)

12. Shovel (**EquipmentID: INT**, Manufacturer: VARCHAR, Shape: VARCHAR, Length: INT)
    a. Manufacturer, Shape, and Length are NOT NULL, FK (EquipmentID) REFERENCES StoresAndCheckedOutEquipment (EquipmentID)

13. Pruner (**EquipmentID: INT**, Manufacturer: VARCHAR, ToolSize: INT, Type: VARCHAR)
    a. Manufacturer, ToolSize, and Type are NOT NULL, FK (EquipmentID) REFERENCES StoresAndCheckedOutEquipment (EquipmentID)

Relationships:

14. Logs (**GardenerID: INT**, **HarvestID: INT**)
    a. PK(GardenerID, HarvestID), FK (GardenerID) REFERENCES Gardener(GardenerID), FK (HarvestID) REFERENCES Harvest(HarvestID)

15. Taken (**HarvestID: INT**, **BedID: INT**, **PlotID: INT**)
    a. PK(HarvestID, BedID, PlotID), FK (HarvestID) REFERENCES Harvest(HarvestID), FK (BedID, PlotID) REFERENCES Bed(BedID, PlotID)

16. PlantedIn (**PlotID: INT**, **BedID: INT**, **Species: VARCHAR**)
    a. PK(BedID, PlotID), FK(PlotID, BedID) REFERENCES Bed(BedID, PlotID), FK(Species) REFERENCES Crop(Species)

17. Records(**Species: VARCHAR**, **PlantingID: INT**)
    a. PK(Species, PlantingID), FK(Species) REFERENCES Crop(Species), FK(PlantingID) REFERENCES PlantingRecord(PlantingID)

18. Rents (**GardenerID: INT**, **PlotID: INT**)
    a. PK(GardenerID, PlotID), FK(GardenerID) REFERENCES Gardener(GardenerID)

19. AssignedTo (**PlotID: INT**, **ScheduleID: INT**)
    a. PK(PlotID, ScheduuleID), FK (ScheduleID) REFERENCES WaterSchedule(ScheduleID), FK (PlotID) REFERENCES Plot(PlotID)

20. Oversees (**ScheduleID: INT**, **GardenerID: INT**)
    a. PK(ScheduleID, GardenerID), FK(ScheduleID) REFERENCES WaterSchedule(ScheduleID), FK(GardenerID) REFERENCES Gardener(GardenerID)

6. Functional Dependencies (FDs)
   a. Identify the functional dependencies in your relations, including the ones involving all candidate keys (including the primary key).

   HarvestYields:
   - HarvestID -> HarvestDate, Weight, Value, Species
   - Species, Weight -> Value

   Gardener:
   - GardenerID -> FirstName, LastName, ResidenceUnitNum, Email
   - Email -> FirstName, LastName
   - ResidenceUnitNum -> LastName

   Crop:
   - Species -> CropName, Growing Season, CropUsage
   - CropName -> CropUsage

WaterSchedule:

- ScheduleID -> StartTime, Temp, Amount, Frequency
- StartTime, Frequency -> Amount

BedContains:

- BedID, PlotID -> FertilizerType, Width, Length, Height

Plot:

- PlotID -> PlotSize, IsAvailable, IsIndoor

ToolShedLocatedAt:

- ShedID -> LastCleanDate, Capacity, NumOfTools, ConditionLevel, PlotID
- LastCleanDate -> ConditionLevel

StoresAndCheckedOutEquipment:

- EquipmentID -> PurchaseDate, EquipmentType, ConditionLevel, ShedID, GardenerID
- PurchaseDate -> ConditionLevel

PlantingRecordTracks:

- PlantingID -> PlantingStyle, PlantingDate, QuantityPlanted, ExpectedHarvestDate, BedID, PlotID

GardenHoe:

- EquipmentID -> LastSharpenDate, ToolSize, PlotsTilled

WateringCan:

- EquipmentID -> MaxVolume, Spout, Material

Shovel:

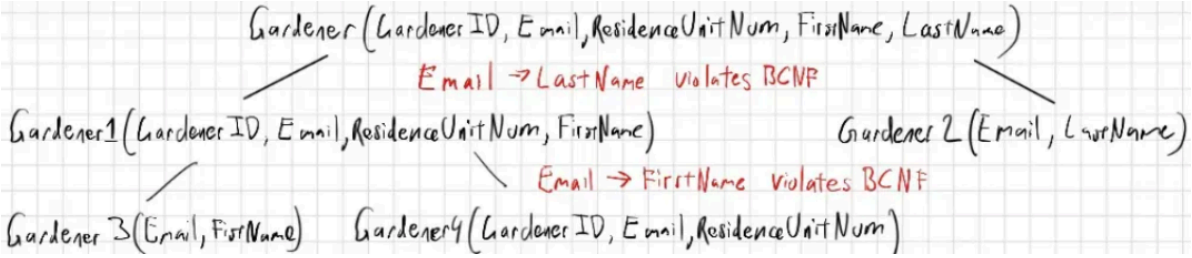
- EquipmentID -> Manufacturer, Shape, Length

Pruner:

- EquipmentID -> Manufacturer, ToolSize, Type

7. Normalization
   a. Normalize each of your tables to be in 3NF or BCNF. Give the list of tables, their primary keys, their candidate keys, and their foreign keys after normalization. You should show the steps taken for the decomposition in a manner similar to that done in class. Should there be errors, and no work is shown, no partial credit can be awarded without steps shown.

   The format should be the same as Step 3, with tables listed similar to Table1(attr1:domain1, attr2:domain2, …). ALL Tables must be listed, not only the ones post normalization.

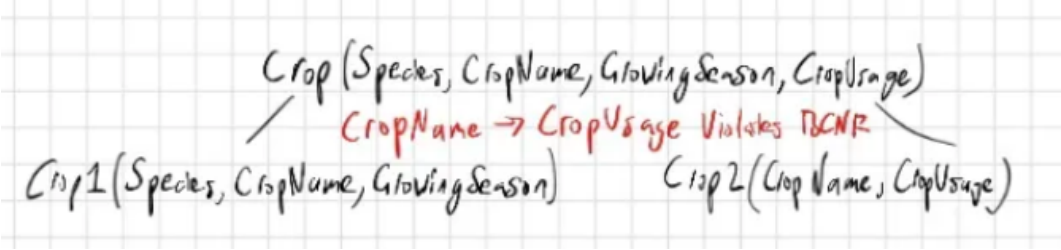


1. Gardener1 (FirstName: VARCHAR, ResidenceUnitNum: INT, GardenerID: INT, **Email: VARCHAR**)
    a. FirstName and ResidenceUnitNum are NOT NULL, Email is UNIQUE, FK(Email) REFERENCES Gardener2 (Email)

2. Gardener2 (LastName: VARCHAR, Email: VARCHAR)
    a. LastName is NOT NULL

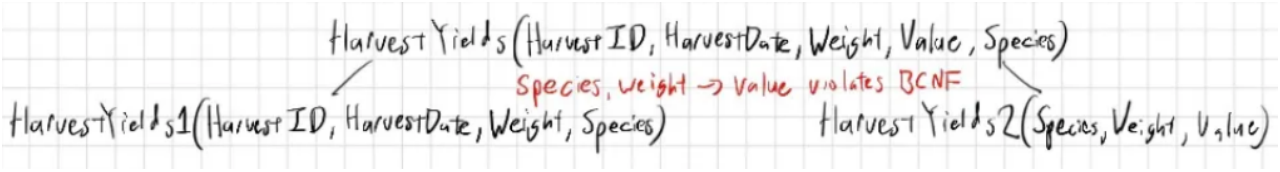
3. Gardener3 (FirstName: VARCHAR, **Email: VARCHAR**)
    a. FirstName is NOT NULL, FK(Email) REFERENCES Gardener2 (Email)

4. Gardener4 (ResidenceUnitNum: INT, GardenerID: INT, **Email: VARCHAR**)
    a. ResidenceUnitNum is NOT NULL, Email is UNIQUE, FK(Email) REFERENCES Gardener2 (Email)



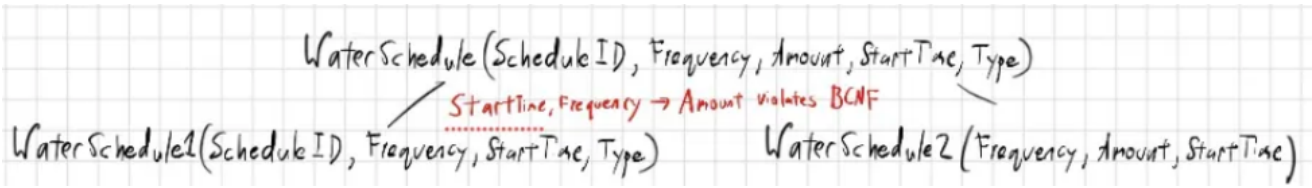
1. Crop1 (Species: VARCHAR, **CropName: VARCHAR**, GrowingSeason: VARCHAR)
    a. CropName and GrowingSeason are NOT NULL, FK (CropName) REFERENCES Crop2 (CropName)

2. Crop2 (CropName: VARCHAR, CropUsage: VARCHAR)



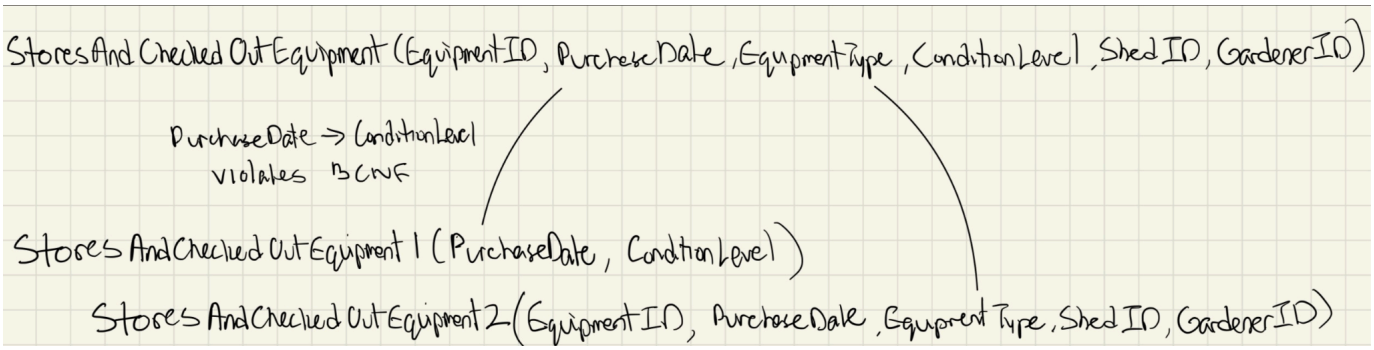
1. HarvestYields1 (HarvestID: INT, HarvestDate: DATE, **Weight: INT**, **Species: VARCHAR**)
    a. HarvestDate, Weight, and Species are NOT NULL, FK (Species) REFERENCES Crop(Species), FK (Species, Weight) REFERENCES HARVESTYields2 (Species, Weight)

2. HarvestYields2 (Weight: INT, Value: INT, **Species: VARCHAR**)
    a. Value is NOT NULL, PK(Weight, Species), FK (Species) REFERENCES Crop(Species)



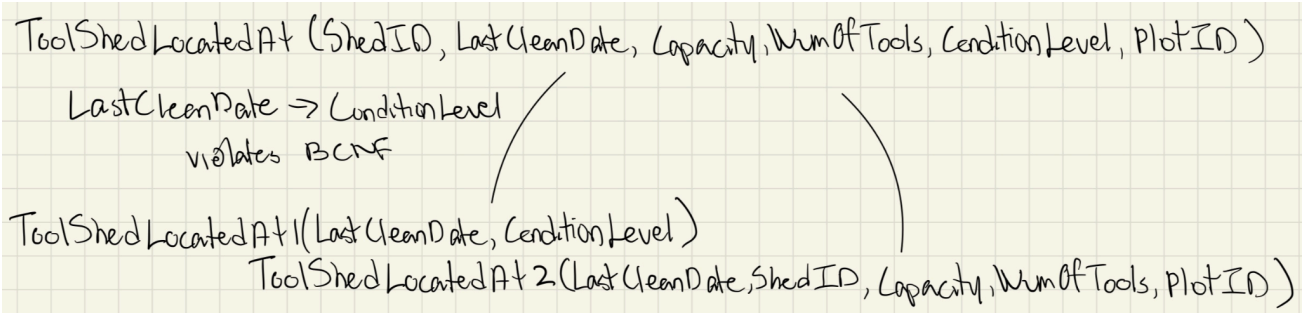
1. WaterSchedule1 (**Frequency: INT**, ScheduleID: INT, **StartTime: VARCHAR**, Temp: INT)
    a. Frequency and StartTime are NOT NULL, FK(StartTime, Frequency) REFERENCES WaterSchedule2 (StartTime, Frequency)

2. WaterSchedule2 (Frequency: INT, Amount: INT, StartTime: VARCHAR)
    a. Amount is NOT NULL, PK(StartTime, Frequency)



1. StoresAndCheckedOutEquipment1 (PurchaseDate: DATE, ConditionLevel: INT)

2. StoresAndCheckedOutEquipment2 (EquipmentID: INT, **PurchaseDate: DATE**, EquipmentType: VARCHAR, **ShedID: INT**, **GardenerID: INT**)
    a. PurchaseDate and EquipmentType are NOT NULL, FK (ShedID) REFERENCES ToolShedLocatedAt2(ShedID), FK (GardenerID) REFERENCES Gardener4(GardenerID), FK (PurchaseDate) REFERENCES StoresAndCheckedOutEquipment1 (PurchaseDate)



1. ToolShedLocatedAt1 (LastCleanDate: DATE, ConditionLevel: INT)

2. ToolShedLocatedAt2 (ShedID: INT, **LastCleanDate: DATE**, Capacity: INT, NumTools: INT, **PlotID: INT**)
    a. Capacity and NumTools are NOT NULL, PlotID is NOT NULL and UNIQUE, FK(PlotID) REFERENCES Plot(PlotID), FK (LastCleanDate) REFERENCES ToolShedLocatedAt1(LastCleanDate)

8. The SQL DDL statements required to create all the tables from item #7.

```
CREATE TABLE Crop2 (
CropName VARCHAR(50)  PRIMARY KEY,
CropUsage VARCHAR(100)
);

CREATE TABLE Crop1 (
Species VARCHAR(50) PRIMARY KEY,
CropName VARCHAR(50)  NOT NULL,
GrowingSeason VARCHAR(20) NOT NULL,
FOREIGN KEY (CropName) REFERENCES Crop2(CropName)
);

CREATE TABLE HarvestYields2 (
Weight INT,
Value INT NOT NULL,
Species VARCHAR(50),
PRIMARY KEY (Weight, Species),
FOREIGN KEY (Species) REFERENCES Crop1(Species)
);

CREATE TABLE HarvestYields1 (
HarvestID INT PRIMARY KEY,
HarvestDate DATE NOT NULL,
Weight INT NOT NULL,
Species VARCHAR(50) NOT NULL,
FOREIGN KEY (Species) REFERENCES Crop1(Species),
FOREIGN KEY (Weight, Species) REFERENCES HarvestYields2(Weight, Species)
);

CREATE TABLE Gardener2 (
LastName VARCHAR(30) NOT NULL,
Email VARCHAR(50) PRIMARY KEY
);
```

```
CREATE TABLE Gardener3 (
FirstName VARCHAR(30) NOT NULL,
Email VARCHAR(50) PRIMARY KEY,
FOREIGN KEY (Email) REFERENCES Gardener2(Email)
);

CREATE TABLE Gardener4 (
ResidenceUnitNum INT NOT NULL,
GardenerID INT PRIMARY KEY,
Email VARCHAR(50) UNIQUE,
FOREIGN KEY (Email) REFERENCES Gardener2(Email)
);

CREATE TABLE Plot (
PlotSize INT,
IsAvailable BOOLEAN NOT NULL,
PlotID INT PRIMARY KEY,
IsIndoor BOOLEAN
);

CREATE TABLE WaterSchedule2 (
Frequency INT,
Amount INT NOT NULL,
StartTime VARCHAR(50),
PRIMARY KEY (StartTime, Frequency)
);

CREATE TABLE WaterSchedule1 (
Frequency INT NOT NULL,
ScheduleID INT PRIMARY KEY,
StartTime VARCHAR(50) NOT NULL,
Temp INT,
FOREIGN KEY (StartTime, Frequency) REFERENCES WaterSchedule2(StartTime,
Frequency)
);

CREATE TABLE BedContains (
Length INT NOT NULL,
Width INT NOT NULL,
```

```
Height INT NOT NULL,
FertilizerType VARCHAR(30),
BedID INT,
PlotID INT,
PRIMARY KEY (BedID, PlotID),
FOREIGN KEY (PlotID) REFERENCES Plot(PlotID)
ON DELETE CASCADE
);

CREATE TABLE PlantingRecordTracks (
PlantedDate DATE,
PlantingID INT PRIMARY KEY,
PlantingStyle VARCHAR(100),
QuantityPlanted INT,
ExpectedHarvestDate DATE,
BedID INT UNIQUE NOT NULL,
PlotID INT UNIQUE NOT NULL,
FOREIGN KEY(BedID, PlotID) REFERENCES BedContains(BedID, PlotID)
);

CREATE TABLE ToolShedLocatedAt1 (
LastCleanDate DATE PRIMARY KEY,
ConditionLevel INT
);

CREATE TABLE ToolShedLocatedAt2 (
ShedID INT PRIMARY KEY,
LastCleanDate DATE,
Capacity INT NOT NULL,
NumTools INT NOT NULL,
PlotID INT UNIQUE NOT NULL,
FOREIGN KEY(PlotID) REFERENCES Plot(PlotID)
ON DELETE CASCADE,
FOREIGN KEY (LastCleanDate) REFERENCES ToolShedLocatedAt1(LastCleanDate)
);

CREATE TABLE StoresAndCheckedOutEquipment1 (
PurchaseDate DATE PRIMARY KEY,
ConditionLevel INT
);
```

```
CREATE TABLE StoresAndCheckedOutEquipment2 (
EquipmentID INT PRIMARY KEY,
PurchaseDate DATE  NOT NULL,
EquipmentType VARCHAR(50)  NOT NULL,
ShedID INT,
GardenerID INT,
FOREIGN KEY (ShedID) REFERENCES ToolShedLocatedAt2(ShedID)
ON DELETE SET NULL,
FOREIGN KEY (GardenerID) REFERENCES Gardener4(GardenerID)
ON DELETE SET NULL,
FOREIGN KEY (PurchaseDate) REFERENCES
StoresAndCheckedOutEquipment1(PurchaseDate)
);

CREATE TABLE GardenHoe (
EquipmentID INT PRIMARY KEY,
PlotsTilled INT,
ToolSize INT NOT NULL,
DateLastSharpened DATE,
FOREIGN KEY (EquipmentID) REFERENCES
StoresAndCheckedOutEquipment2(EquipmentID)
ON DELETE CASCADE
);

CREATE TABLE WateringCan (
EquipmentID INT PRIMARY KEY,
MaxVolume INT NOT NULL,
Spout VARCHAR(30) NOT NULL,
Material VARCHAR(30) NOT NULL,
FOREIGN KEY (EquipmentID) REFERENCES
StoresAndCheckedOutEquipment2(EquipmentID)
ON DELETE CASCADE
);

CREATE TABLE Shovel (
EquipmentID INT PRIMARY KEY,
Manufacturer VARCHAR(30) NOT NULL,
Shape VARCHAR(30) NOT NULL,
Length INT NOT NULL,
```

```
FOREIGN KEY (EquipmentID) REFERENCES
StoresAndCheckedOutEquipment2(EquipmentID)
ON DELETE CASCADE
);

CREATE TABLE Pruner (
EquipmentID INT PRIMARY KEY,
Manufacturer VARCHAR(30) NOT NULL,
ToolSize INT NOT NULL,
Type VARCHAR(30) NOT NULL,
FOREIGN KEY (EquipmentID) REFERENCES
StoresAndCheckedOutEquipment2(EquipmentID)
ON DELETE CASCADE
);


CREATE TABLE Logs (
GardenerID INT,
HarvestID INT,
PRIMARY KEY(GardenerID, HarvestID),
FOREIGN KEY (GardenerID) REFERENCES Gardener4(GardenerID)
ON DELETE CASCADE,
FOREIGN KEY (HarvestID) REFERENCES HarvestYields1(HarvestID)
ON DELETE CASCADE
);

CREATE TABLE Taken (
HarvestID INT,
BedID INT,
PlotID INT,
PRIMARY KEY(HarvestID, BedID, PlotID),
FOREIGN KEY (HarvestID) REFERENCES HarvestYields1(HarvestID)
ON DELETE CASCADE,
FOREIGN KEY (BedID, PlotID) REFERENCES BedContains(BedID, PlotID)
ON DELETE CASCADE
);

CREATE TABLE PlantedIn (
BedID INT,
PlotID INT,
```

```
Species VARCHAR(50),
PRIMARY KEY(BedID, PlotID),
FOREIGN KEY(BedID, PlotID) REFERENCES BedContains(BedID, PlotID)
ON DELETE CASCADE,
FOREIGN KEY(Species) REFERENCES Crop1(Species)
ON DELETE CASCADE
);

CREATE TABLE Records(
Species VARCHAR(50),
PlantingID INT,
PRIMARY KEY(Species, PlantingID),
FOREIGN KEY(Species) REFERENCES Crop1(Species)
ON DELETE CASCADE,
FOREIGN KEY(PlantingID) REFERENCES PlantingRecordTracks(PlantingID)
ON DELETE CASCADE
);

CREATE TABLE Rents (
GardenerID INT,
PlotID INT,
PRIMARY KEY(GardenerID, PlotID),
FOREIGN KEY(GardenerID) REFERENCES Gardener4(GardenerID)
ON DELETE CASCADE
);

CREATE TABLE AssignedTo (
PlotID INT,
ScheduleID INT,
PRIMARY KEY (PlotID, ScheduleID),
FOREIGN KEY (ScheduleID) REFERENCES WaterSchedule1(ScheduleID)
ON DELETE CASCADE,
FOREIGN KEY (PlotID) REFERENCES Plot(PlotID)
ON DELETE CASCADE
);

CREATE TABLE Oversees (
ScheduleID INT,
GardenerID INT,
PRIMARY KEY(ScheduleID, GardenerID),
```

```
FOREIGN KEY(ScheduleID) REFERENCES WaterSchedule1(ScheduleID)
ON DELETE CASCADE,
FOREIGN KEY(GardenerID) REFERENCES Gardener4(GardenerID)
ON DELETE CASCADE
);
```

```
INSERT INTO  Crop2
VALUES ('Carrot', 'Juicing'),
        ('Lettuce', 'Salads'),
        ('Corn', 'Grilled corn'),
        ('Tomato', 'Tomato sauces'),
        ('Kale', 'Smoothies');

INSERT INTO  Crop1
VALUES ( 'Daucus carota', 'Carrot', 'Spring'),
        ('Lactuca sativa', 'Lettuce', 'Spring, Summer'),
        ('Zea mays', 'Corn', 'Summer, Autumn'),
        ('Solanum lycopersicum',  'Tomato', 'Summer'),
        ('Brassica oleracea', 'Kale', 'Winter');

INSERT INTO HarvestYields2
VALUES (1500, 1200,  'Daucus carota'),
        (500, 850, 'Lactuca sativa'),
        (1800, 1100, 'Zea mays'),
        (2000, 1800, 'Solanum lycopersicum'),
        (400, 750, 'Brassica oleracea');

INSERT INTO HarvestYields1
VALUES (201, DATE '2026-05-10', 1500, 'Daucus carota'),
        (202, DATE '2026-06-14', 500, 'Lactuca sativa'),
        (203, DATE '2026-08-10', 1800, 'Zea mays'),
        (204, DATE '2026-07-22', 2000, 'Solanum lycopersicum'),
        (205, DATE '2026-12-05', 400, 'Brassica oleracea');

INSERT INTO Gardener2
VALUES ('Wang', 'wong@gmail.com'),
        ('Smith', 'smith@gmail.com'),
        ('Singh', 'singh@gmail.com'),
        ('Kumar', 'kumar@outlook.com'),
        ('Ahmed', 'ahmed@hotmail.com');

INSERT INTO Gardener3
VALUES ('David', 'wong@gmail.com'),
        ('Lacy', 'smith@gmail.com'),
        ('Sam', 'singh@gmail.com'),
        ('George', 'kumar@outlook.com'),
```

```
        ('Adrian', 'ahmed@hotmail.com');

INSERT INTO Gardener4
VALUES (0, 1, 'wong@gmail.com'),
        (1, 2, 'smith@gmail.com'),
        (2, 3, 'singh@gmail.com'),
        (3, 4, 'kumar@outlook.com'),
        (4, 5, 'ahmed@hotmail.com');

INSERT INTO Plot
VALUES (0, TRUE, 1, TRUE),
        (1, FALSE, 2, TRUE),
        (10, TRUE, 3, FALSE),
        (15, FALSE, 4, FALSE),
        (230, FALSE, 5, TRUE);

INSERT INTO WaterSchedule2
VALUES (3, 25, '10am'),
        (4, 15, '11am'),
        (6, 20, '12pm'),
        (6, 40, '6am'),
        (4, 11, '7am');

INSERT INTO WaterSchedule1
VALUES (3, 1, '10am', 24),
        (4, 2, '11am', 16),
        (6, 3, '12pm', 18),
        (6, 4, '6am', 25),
        (4, 5, '7am', 22);

INSERT INTO BedContains
VALUES (1, 1, 1, 'Organic', 1, 1),
        (3, 2, 6, 'Synthetic', 2, 2),
        (9, 5, 4, 'Organic', 3, 3),
        (6, 7, 6, 'Biofertilizer', 4, 4),
        (2, 5, 1, 'Synthetic', 5, 5);

INSERT INTO PlantingRecordTracks
VALUES (DATE '2026-07-30', 1, 'Hill', 10, DATE '2026-09-30', 1, 1),
        (DATE '2026-08-01', 2, 'Row', 12, DATE '2026-09-01', 2, 2),
        (DATE '2026-03-17', 3, 'Strip', 1, DATE '2026-04-15', 3, 3),
        (DATE '2026-05-24', 4, 'Broadcast', 23, DATE '2026-10-23', 4, 4),
        (DATE '2026-03-10', 5, 'Raised Bed', 5, DATE '2026-04-17', 5, 5);

INSERT INTO ToolShedLocatedAt1
VALUES (DATE '2026-07-01', 4),
```

```
        (DATE '2026-04-01', 1),
        (DATE '2026-06-15', 3),
        (DATE '2026-07-15', 5),
        (DATE '2025-05-15', 2);

INSERT INTO ToolShedLocatedAt2
VALUES (1, DATE '2026-07-01', 30, 18, 1),
        (2, DATE '2026-06-15', 25, 15, 2),
        (3, DATE '2026-07-15', 80, 45, 3),
        (4, DATE '2025-05-15', 45, 23, 4),
        (5, DATE '2026-04-01', 10, 6, 5);

INSERT INTO StoresAndCheckedOutEquipment1
VALUES (DATE '2026-03-15', 2),
        (DATE '2026-04-15', 4),
        (DATE '2026-05-15', 5),
        (DATE '2026-05-22', 4),
        (DATE '2026-06-22', 5);

INSERT INTO StoresAndCheckedOutEquipment2
VALUES (101, DATE '2026-03-15', 'Planting', 1, 1),
        (102, DATE '2026-05-22', 'Irrigation', 1, 1),
        (103, DATE '2026-04-15', 'Harvesting', 2, NULL),
        (601, DATE '2026-05-15', 'Maintenance', 4, 3),
        (202, DATE '2026-06-22', 'Transportation', 5, NULL);
```

ISA (Corresponding EquipmentID in StoresAndCheckedOutEquipment2 may not exist):

```
    INSERT INTO GardenHoe
    VALUES (601, 12, 2, DATE '2026-04-30'),
            (602, 16, 3, DATE '2026-05-12'),
            (603, 43, 6, DATE '2026-04-15'),
            (604, 0, 6, DATE '2026-05-22'),
            (607, 67, 7, DATE '2026-06-07');

    INSERT INTO WateringCan
    VALUES (401, 10, 'Rose shower', 'Plastic' ),
            (402, 15, 'Rose shower', 'Steel'),
            (403, 5, 'Narrow stream', 'Plastic'),
            (404, 5, 'Rose shower', 'Plastic'),
            (405, 10, 'Narrow stream', 'Steel');

    INSERT INTO Shovel
    VALUES (101, 'Fiskars', 'Round Point', 48),
            (102, 'Garant', 'Scoop', 44),
            (103, 'True Temper', 'Square Point', 46),
```

```
        (104, 'Bully Tools', 'Trenching', 21),
        (105, 'AMES', 'Drain', 1);

INSERT INTO Pruner
VALUES (301, 'Felco', 8, 'Bypass'),
        (302, 'Corona', 11, 'Anvil'),
        (303, 'Felco', 10, 'Anvil'),
        (304, 'Okatsune', 5, 'Floral snips'),
        (305, 'Corona', 6, 'Bypass');
```

Relationships:

```
INSERT INTO Logs
VALUES (1, 201),
       (2, 202),
       (3, 203),
       (4, 204),
       (5, 205);

INSERT INTO Taken
VALUES (201,1, 1),
       (202, 2, 2),
       (202, 3, 3),
       (204, 4, 4),
       (205, 5, 5);

INSERT INTO PlantedIn
VALUES (1, 1, 'Daucus carota'),
       (2, 2, 'Lactuca sativa'),
       (3, 3, 'Zea mays'),
       (4, 4, 'Solanum lycopersicum'),
       (5, 5, 'Brassica oleracea');

INSERT INTO Records
VALUES ('Daucus carota', 1),
       ('Zea mays', 2),
       ('Zea mays', 3),
       ('Solanum lycopersicum', 4),
       ('Brassica oleracea', 5);

INSERT INTO Rents
VALUES (1, 1),
       (2, 4),
       (3, 3),
       (4, 2),
       (5, 5);
```

```
INSERT INTO AssignedTo
VALUES (1, 2),
       (4, 5),
       (2, 3),
       (3, 4),
       (5, 1);

INSERT INTO Oversees
VALUES (1, 5),
       (2, 4),
       (3, 3),
       (4, 2),
       (5, 1);
```

9. An explicit acknowledgment about your use of AI tools in this assignment. Specifically, we are looking for a clear yes/no about whether you have used one or more AI tools. If yes, we want to know which tool(s) you have used, and we want a PDF of the full conversation.

No AI used.